



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

HYGIENE AND SAFETY PRECAUTIONS MEASURES

LSVHS302

Apply hygiene and safety precautions measures

Competence

RQF Level: 3

Learning Hours



Credits: 3

Sector: Construction and Building Services

Trade: Land Surveying

Module Type: Specific

Curriculum: CBSLSV3001- TVET Certificate 3 in Land Surveying

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Purpose statement	This module is designed for a student pursuing a level 3 TVET certificate in Land Surveying. It describes the skills, knowledge, and right attitudes required to apply hygiene and safety precautions measures at surveying the workplace. At the end of this module, the students will be able to select surveying personal protective equipment, identify and handle hazards and emergencies at the surveying workplace, and to maintain surveying equipment.					
Leaning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	100%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	20%				
			Summative assessment		N.A.	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Select surveying personal	1.1. Types of surveying personal protective equipment are properly identified according to the weather condition, type of work,

protective equipment	and terrain characteristics.
	1.2. Surveying personal protective equipment is appropriately used according to the weather condition, type of work, and terrain characteristics.
	1.3. Personal protective equipment is properly cared according to their maintenance and storage system.
2. Identify and handle Hazards and emergencies at surveying the Workplace	2.1. Hazards and emergencies are properly identified at surveying workplace.
	2.2. Surveying hazards and emergencies are appropriately handled at the workplace.
	2.3. Hazards and emergencies are adequately evaluated according to their types.
3. Maintain surveying equipment	3.1. Hygiene of surveying equipment is appropriately done according to the types of equipment and manufacturer's catalogue
	3.2. Surveying equipment is appropriately transported according to the type of equipment
	3.3. Surveying equipment is adequately stored according to manufacturer's catalogue

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Select surveying personal protective equipment 2. Identify and handle hazards and emergencies in surveying the workplace 3. Maintain surveying instruments
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Learning outcome 1: Select surveying personal protective equipment		Learning hours: 7
Indicative content		
* Types of surveying personal protective equipment (Refer to RS ISO 20347, RS ISO 13688, ISO 3873, ISO 21420, ISO 17420-1) - Head protection, foot protection, hands protection, body protection, eye protection, ear protection, respiratory protection, etc. * The use of surveying personal protective equipment (PPE) - Helmet, safety shoes, safety boots, gloves, overalls, reflector jacket, safety goggles, safety glasses, earplugs, earmuff, mask, etc. * Maintenance and storage of surveying personal protective equipment - Maintenance of PPE (cleaning, examination, replacement, repair, and testing) - Storage of PPE (Hanging, Laying, clothing, container/box)		
Resources required for the learning outcome		
Equipment	High visibility jackets, safety boots & knee pads, gloves, glasses & goggles, respiratory masks, earplugs & earmuff, and safety helmets, whiteboard, projector, computer	
Materials	Water, pens, notebooks, marker pen, soap, flipcharts, etc.	
Tools	Surveyor's bags, brushes, sponge, etc....	
Facilitation techniques	- Group work - Practical exercise - Discussion - Brainstorming - Trainer guided	
Formative assessment methods	- Written assessment - Oral assessment	

Learning outcome 2: Identify Hazards and emergencies at surveying Workplace		Learning hours: 9
Indicative content		
* Identify Hazards and emergencies at surveying the Workplace (Refer to ISO 18400-103) - Types of hazards (Safety hazards, Chemical hazards, Biological hazards). - Types of emergencies (Natural emergencies, Work-related emergencies). - Comparison between hazard and emergency. * Handling surveying hazards and emergencies at the workplace - Method of handling surveying hazards at the workplace (Use protective clothing and equipment, Improve work policies and procedures). - Method of handling surveying emergencies at the workplace (Determine the nature of the emergency, remain calm, assess the immediate threat, etc.). * Evaluation of Hazards and emergencies - Steps/procedures to evaluate types of hazards at surveying workplace. - Steps/procedures to evaluate types of emergencies at surveying workplace.		
Resources required for the indicative content		
Equipment	Measuring devices, fire extinguishers, first aid kits, whiteboard, flipcharts stand, projector, computer.	
Materials	Chalk/paint, pens, notebooks, marker pen, flipcharts.	
Tools	Safety cones, videos, stories and pictures	
Facilitation techniques	- Demonstration and simulation - Individual and group work - Practical exercise - Case study - Trainer guided - Group discussion	
Formative assessment	Written assessment	

methods	• Oral assessment
Learning outcome 3: Maintain surveying equipment	
Learning hours: 14	
Indicative content	
* Hygiene of surveying equipment (Refer to ISO 9849, ISO 12510) • Cleaning method, • Lubrication methods * Transportation of surveying equipment • Operator's Manual, • Routine Care of Equipment * Storage of surveying equipment • Storeroom • Casing and uncasing	
Resources required for the indicative content	
Equipment	Storeroom, vehicle, computer, dumpy level, theodolite, total station, GPS, whiteboard, computer, and projector.
Materials	Water, oil, brushes, marker pen, pieces of chalk, notebook.
Tools	Toolbox, markers, wheelbarrow
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment

References:

1. Hugo, C, .(April 2012). The calibration of survey equipment.
2. Safety manual - Division of Plats and Survey, Maryland State Highway Administration.
3. Goma Epidemiology Group. (1995). Public health in disasters and emergencies.
4. Perrin. (1996). Médecins sans Frontières (1997a).
5. International Federation of Red Cross and Red Crescent Societies (2000).
6. International Federation of Red Cross and Red Crescent Societies. (2000) Johns Hopkins University
7. Work Readiness Training Programme-Trainer's Manual, Akazi Kanoze-Youth Livelihoods Project.
8. RS ISO 20347 Personal protective equipment — Occupational footwear
9. RS ISO 13688 /AMD 1 Protective clothing — General requirements — Amendment 1
10. ISO 3873 Industrial Safety Helmets
11. ISO 21420/AMD 1 Protective Gloves — General Requirements And Test Methods — Amendment 1
12. ISO 17420-1 Respiratory Protective Devices — Performance Requirements — Part 1: General
13. ISO 9849 Optics And Optical Instruments — Geodetic And Surveying Instruments — Vocabulary
14. ISO 12510 Earth-Moving Machinery — Operation And Maintenance — Maintainability Guidelines
15. ISO 18400-103 Soil Quality — Sampling — Part 103: Safety